Real- time Ultrasound Volterra Filtering on FPGA

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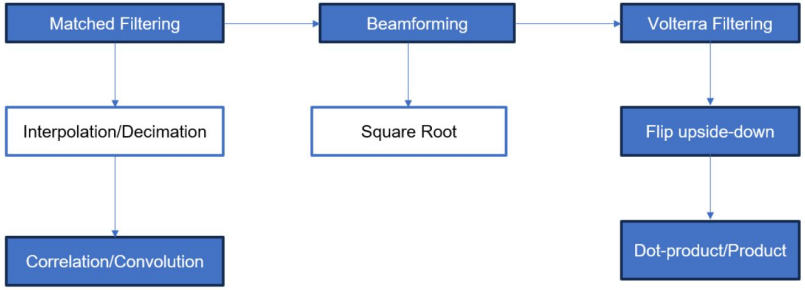
Project Overview

Goal : To implement a second-order volterra filter for ultrasound non-linear imaging on FPGA

Approach : Translate reference MATLAB functions in Verilog code to represent it on FPGA. These FPGAs provide the necessary parallel processing and speed for real-time ultrasound signal processing compared to software alone.

Functions



My roles and objectives

Assigned function: Matched Filtering - Convolution and Interpolation functions.

Key Responsibilities:

- Designing the RTL architecture in Verilog
- Developing Testbench for each function to calculate and analyse the error wrt. MATLAB software output.

Data Size:

- The size of data is 1792x128x100 where 1792 is the number of samples acquired, 128 is the number of transducer elements and 100 frames are acquired.

Weekly Progress : Week 1 & 2

- Overviewing the existing paper-work on Volterra filtering and its application in ultrasound imaging. Paper
- Learning about the overall project - Contrast-Enhanced Imaging of Microvessel imaging Using Chirp-Coded Excitation and Volterra Filtering
  More about the overall project

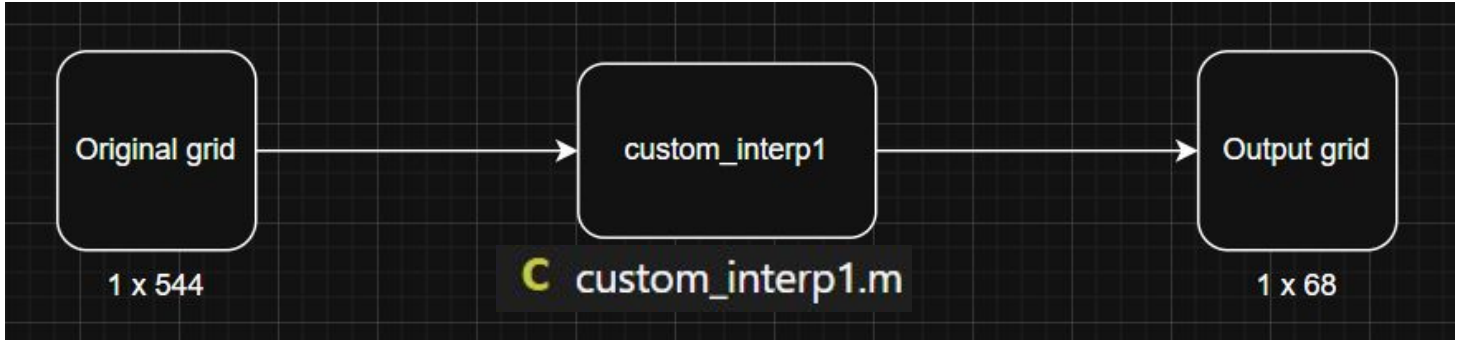
- Executing the provided MATLAB code to understand the expected inputs, outputs, and different data sizes. `Volterra_code_with_custom.m`
- Understanding how to implement negative and floating point values in Verilog

Weekly Progress : Week 3 & 4

- I was assigned Interpolation/Decimation function: `custom_interp1`
- Then I understood what interpolation and decimation mean for signal preprocessing and came to the conclusion that decimation is required here with no need of an aliasing filter.
- Designed the RTL code and testbench for the function.

Interpolation Function

```
ww1 = custom_interp1(t_orig1, tw1, t_new1)
```



- t_orig1 : 1x544 step size grid (initial sampling freq., step size = 4e -09)
- tw1 : original input grid (544 data points)
- t_new1 : array of 1x68 (stepsize = 3.2 e-08) that shows the new sampling points
- ww1 : output grid (68 new points)

[MATLAB] : vq = custom_interp1(x,y,xq)

According to the MATLAB code , we need to down sample the original signals according to the interpolation relation

$$y = y1 * t + y2 * (1-t) ;$$

rearranging this;

$$y = y2 + t(y1-y2);$$

where t = (current_xq - x1)/(x2-x1);

For it's FPGA implementation,

- Floating point conversions to fixed RTL code
- Using shift register and multiplication accumulator logic to speed up
- Using decimal format for data (was in double format- floating point)
- Testbench to calculate the error and run synthesis

Challenges faced and solutions

- Floating point values : RTL has fixed point interpretation, so I used q2.30 decimal format to incorporate both overflow and floating point values
- Overflow : Initially using 16 bit for data values and 32 bit for multiplication led to overflow so I switched to 32 bit data width and 64 bit multiplication.
- High error : Vivado didn’t execute the whole code, so I had to run several iterations, reduce latency and using the run -all function , the error reduced.
- Normalisation : To test the MATLAB values, normalisation was required.

interp.v | tb.v | Untitled 1

Name	Value
clk	1
rst	0
y_in[31:0]	0
y_in_valid	0
vq_out[31:0]	0
vq_out_valid	0
y_input_real[0:543]	0.0,0.0,0.0,-0.008443705,
vq_expected_real[0:67]	-0.201246277,-0.2444003
rtl_outputs[0:67]	-0.201246277000789,-0.2
input_index[31:0]	544
output_index[31:0]	68
k[31:0]	68
rtl_max	1.13159656710923
error	0.0
max_error	1.53531374419424e-09
DATA_WIDTH[31:0]	32
SCALE	1073741824.0
TOL	0.001

/tb_custom_interp1/clk

2,704.000 ns

0.000 ns | 500.000 ns | 1,000.000 ns | 1,500.000 ns | 2,000.000 ns | 2,500.0

0

0

0.0,0.0,0.0,-0.008443705,-0.039431026,-0.0931578,-0.159089109,-0.227729597,-0.291070076,-0.3428602

-0.201246277,-0.24440033,0.63511929,0.592471971,-0.610796492,-0.947357699,0.28647898,1.0,-0.043538

-0.2277295961976

544

68

X

0.0

0.0

0.0

32

1073741824.0

0.001

Synthesis using Nexys4

SYNTHESIZED DESIGN - xc7a100tcsg324-1

Sources | Netlist

custom_interp1
- Nets (626)
  - current_time_reg (16)
  - diff (33)
  - frac_32 (2)
  - p_0_in (32)
  - target_time_reg (13)
  - vq_out (32)
    - vq_out[0]
    - vq_out[1]
    - vq_out[2]
    - vq_out[3]
    - vq_out[4]
    - vq_out[5]

Bus Net Properties

vq_out

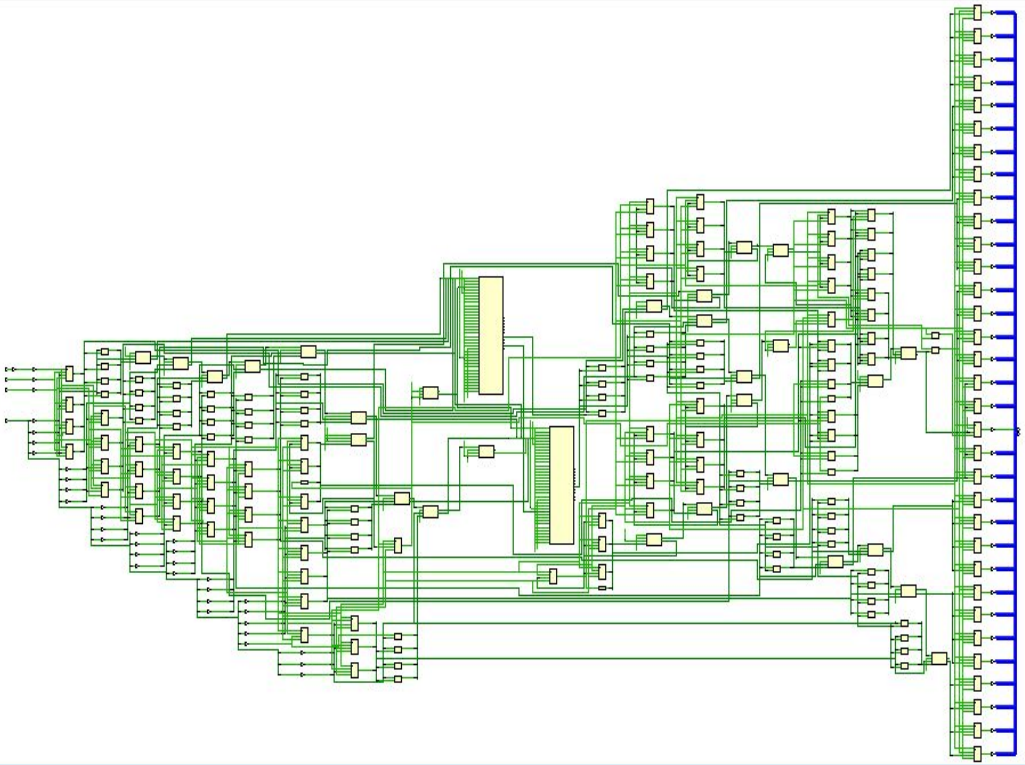
Name: vq_out

Number of nets: 32

General | Scalar Nets

Project Summary | Device | interp.v | tb.v | Schematic

277 Cells 68 I/O Ports 626 Nets



Results and error (RTL vs MATLAB)

The output very closely matches the MATLAB data output.

Error is in order of 1e -08.

```
=====================================
Comparing RTL vs MATLAB
=====================================
xq=0 RTL=-0.2012462770 MATLAB=-0.2012462770 ERROR=0.0000000000
xq=1 RTL=-0.2444003299 MATLAB=-0.2444003300 ERROR=0.0000000001
xq=2 RTL=0.6351192906 MATLAB=0.6351192900 ERROR=0.0000000006
xq=3 RTL=0.5924719714 MATLAB=0.5924719710 ERROR=0.0000000004
xq=4 RTL=-0.6107964932 MATLAB=-0.6107964920 ERROR=0.0000000012
xq=5 RTL=-0.9473576992 MATLAB=-0.9473576990 ERROR=0.0000000002
xq=6 RTL=0.2864789801 MATLAB=0.2864789800 ERROR=0.0000000001
xq=7 RTL=1.0000000000 MATLAB=1.0000000000 ERROR=0.0000000000
xq=8 RTL=-0.0435383720 MATLAB=-0.0435383730 ERROR=0.0000000010
xq=9 RTL=-0.9457512279 MATLAB=-0.9457512270 ERROR=0.0000000009
xq=10 RTL=-0.1040767179 MATLAB=-0.1040767180 ERROR=0.0000000001
xq=11 RTL=0.9695663715 MATLAB=0.9695663710 ERROR=0.0000000005
xq=12 RTL=0.1472010748 MATLAB=0.1472010750 ERROR=0.0000000002
xq=13 RTL=-0.9839901394 MATLAB=-0.9839901390 ERROR=0.0000000004
xq=14 RTL=-0.1559056875 MATLAB=-0.1559056880 ERROR=0.0000000005
xq=15 RTL=0.9816821956 MATLAB=0.9816821950 ERROR=0.0000000006
xq=16 RTL=0.1500697751 MATLAB=0.1500697750 ERROR=0.0000000001
xq=17 RTL=-0.9854808518 MATLAB=-0.9854808510 ERROR=0.0000000008
xq=18 RTL=-0.0893317102 MATLAB=-0.0893317100 ERROR=0.0000000002
xq=19 RTL=0.9743691711 MATLAB=0.9743691710 ERROR=0.0000000001
xq=20 RTL=-0.0775453058 MATLAB=-0.0775453060 ERROR=0.0000000002
xq=21 RTL=-0.9354166295 MATLAB=-0.9354166290 ERROR=0.0000000005
xq=22 RTL=0.3660391816 MATLAB=0.3660391820 ERROR=0.0000000004
xq=23 RTL=0.7684092708 MATLAB=0.7684092710 ERROR=0.0000000002
xq=24 RTL=-0.6850677326 MATLAB=-0.6850677330 ERROR=0.0000000004
xq=25 RTL=-0.4353823673 MATLAB=-0.4353823670 ERROR=0.0000000003
xq=26 RTL=0.8901385530 MATLAB=0.8901385520 ERROR=0.0000000010
xq=27 RTL=0.0535589625 MATLAB=0.0535589630 ERROR=0.0000000005
xq=28 RTL=-0.9208709448 MATLAB=-0.9208709440 ERROR=0.0000000008
xq=29 RTL=0.4064952763 MATLAB=0.4064952770 ERROR=0.0000000007
xq=30 RTL=0.6835057009 MATLAB=0.6835057000 ERROR=0.0000000009
xq=31 RTL=-0.8237814208 MATLAB=-0.8237814200 ERROR=0.0000000008
xq=32 RTL=-0.1032978843 MATLAB=-0.1032978850 ERROR=0.0000000007
xq=33 RTL=0.8982846627 MATLAB=0.8982846620 ERROR=0.0000000007
xq=34 RTL=-0.5466146403 MATLAB=-0.5466146410 ERROR=0.0000000007
```

Results and error (RTL vs MATLAB)

```
xq=33 RTL=0.8982846627 MATLAB=0.8982846620 ERROR=0.0000000007
xq=34 RTL=-0.5466146403 MATLAB=-0.5466146410 ERROR=0.0000000007
xq=35 RTL=-0.5193113523 MATLAB=-0.5193113520 ERROR=0.0000000003
xq=36 RTL=0.9005752911 MATLAB=0.9005752910 ERROR=0.0000000001
xq=37 RTL=-0.1229455952 MATLAB=-0.1229455960 ERROR=0.0000000008
xq=38 RTL=-0.7735563579 MATLAB=-0.7735563580 ERROR=0.0000000001
xq=39 RTL=0.7349506222 MATLAB=0.7349506220 ERROR=0.0000000002
xq=40 RTL=0.0293633599 MATLAB=0.0293633610 ERROR=0.0000000011
xq=41 RTL=-0.7300384898 MATLAB=-0.7300384890 ERROR=0.0000000008
xq=42 RTL=0.7216643938 MATLAB=0.7216643940 ERROR=0.0000000002
xq=43 RTL=0.0258913309 MATLAB=0.0258913320 ERROR=0.0000000011
xq=44 RTL=-0.6850862176 MATLAB=-0.6850862170 ERROR=0.0000000006
xq=45 RTL=0.7014039242 MATLAB=0.7014039240 ERROR=0.0000000002
xq=46 RTL=-0.2543803680 MATLAB=-0.2543803690 ERROR=0.0000000010
xq=47 RTL=-0.3290938034 MATLAB=-0.3290938040 ERROR=0.0000000006
xq=48 RTL=0.7001865569 MATLAB=0.7001865570 ERROR=0.0000000001
xq=49 RTL=-0.5777925498 MATLAB=-0.5777925490 ERROR=0.0000000008
xq=50 RTL=0.0582907745 MATLAB=0.0582907750 ERROR=0.0000000005
xq=51 RTL=0.4934350580 MATLAB=0.4934350580 ERROR=0.0000000000
xq=52 RTL=-0.7145354485 MATLAB=-0.7145354480 ERROR=0.0000000005
xq=53 RTL=0.4799232951 MATLAB=0.4799232950 ERROR=0.0000000001
xq=54 RTL=0.1313558734 MATLAB=0.1313558740 ERROR=0.0000000006
xq=55 RTL=-0.3831918490 MATLAB=-0.3831918490 ERROR=0.0000000000
xq=56 RTL=-0.0232307302 MATLAB=-0.0232307310 ERROR=0.0000000008
xq=57 RTL=0.0948946569 MATLAB=0.0948946570 ERROR=0.0000000001
xq=58 RTL=0.0419788002 MATLAB=0.0419788010 ERROR=0.0000000008
xq=59 RTL=0.0284167683 MATLAB=0.0284167690 ERROR=0.0000000007
xq=60 RTL=0.0099196270 MATLAB=0.0099196270 ERROR=0.0000000000
xq=61 RTL=-0.0160535315 MATLAB=-0.0160535330 ERROR=0.0000000015
xq=62 RTL=-0.0175656917 MATLAB=-0.0175656920 ERROR=0.0000000003
xq=63 RTL=-0.0055697999 MATLAB=-0.0055698000 ERROR=0.0000000001
xq=64 RTL=-0.0009638304 MATLAB=-0.0009638310 ERROR=0.0000000006
xq=65 RTL=0.0044398609 MATLAB=0.0044398610 ERROR=0.0000000001
xq=66 RTL=0.0054012873 MATLAB=0.0054012880 ERROR=0.0000000007
xq=67 RTL=0.0000000000 MATLAB=0.0000000000 ERROR=0.0000000000

====================================
MAX ABS ERROR = 0.000000
RESULT: PASS
====================================
```

Week 5 & 6

- I was assigned Convolution/Correlation function: `custom_conv`
- Then I understood that here we need to perform a cross correlation and don't need to reverse the output values. Hence, I came to the conclusion that no flipping is required in the end and cross correlation would work.
- I also figured that here the kernel is a 1D array and we only need to iterate through it. This enabled us faster computation compared to 2D convolution.
- Designed the RTL code and testbench for the function.

Convolution Function

```
x2_conv_orig = custom_conv(x', FW2);
```

- x2_conv_orig : output 128 x 1792 grid of data
- x’ : transposed data as input
- Fw2 : kernel length 1D array of 136 values (slides across to calculate)

Also, padding of 0’s was needed both sides to perform sliding convolution.

MATLAB : output_matrix = custom_conv(input1, input2)

- For sliding kernel values, 136-tap architecture is used :

(shift registers -> parallel multipliers -> accumulation tree -> output value)

- The 128 columns were streamed sequentially as individual rows. Each row was padded with 68 zeros on the left and 67 zeros on the right, making each streaming cycle exactly 1927 clock cycles long.

- Used .mem files for smooth and faster data read

csv_to_mem

$$Y[n] = \sum_{i=0}^{TAPS-1} \text{data}[n-i] \times \text{coeff}[i]$$

Challenges faced and solutions

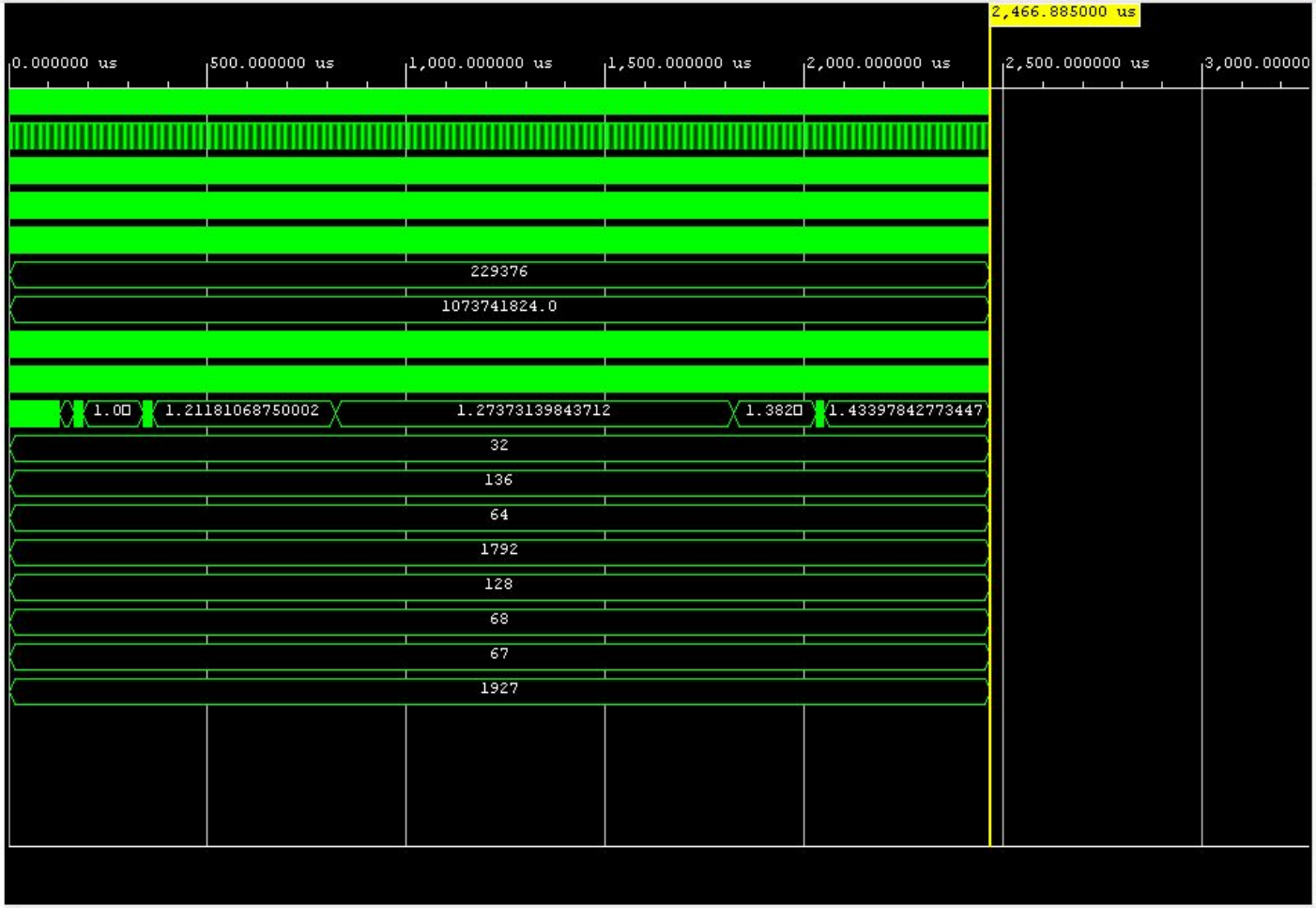
- Floating point values : RTL has fixed point interpretation, so I used 32 bit data width and decimal format to incorporate both overflow and floating point values

- Time limit : For faster multiplication, used a tree like architecture to calculate the sliding kernel, then use parallel multiplication and evaluate using adders.

- No need of reversing the grid as cross- correlation is sufficient.

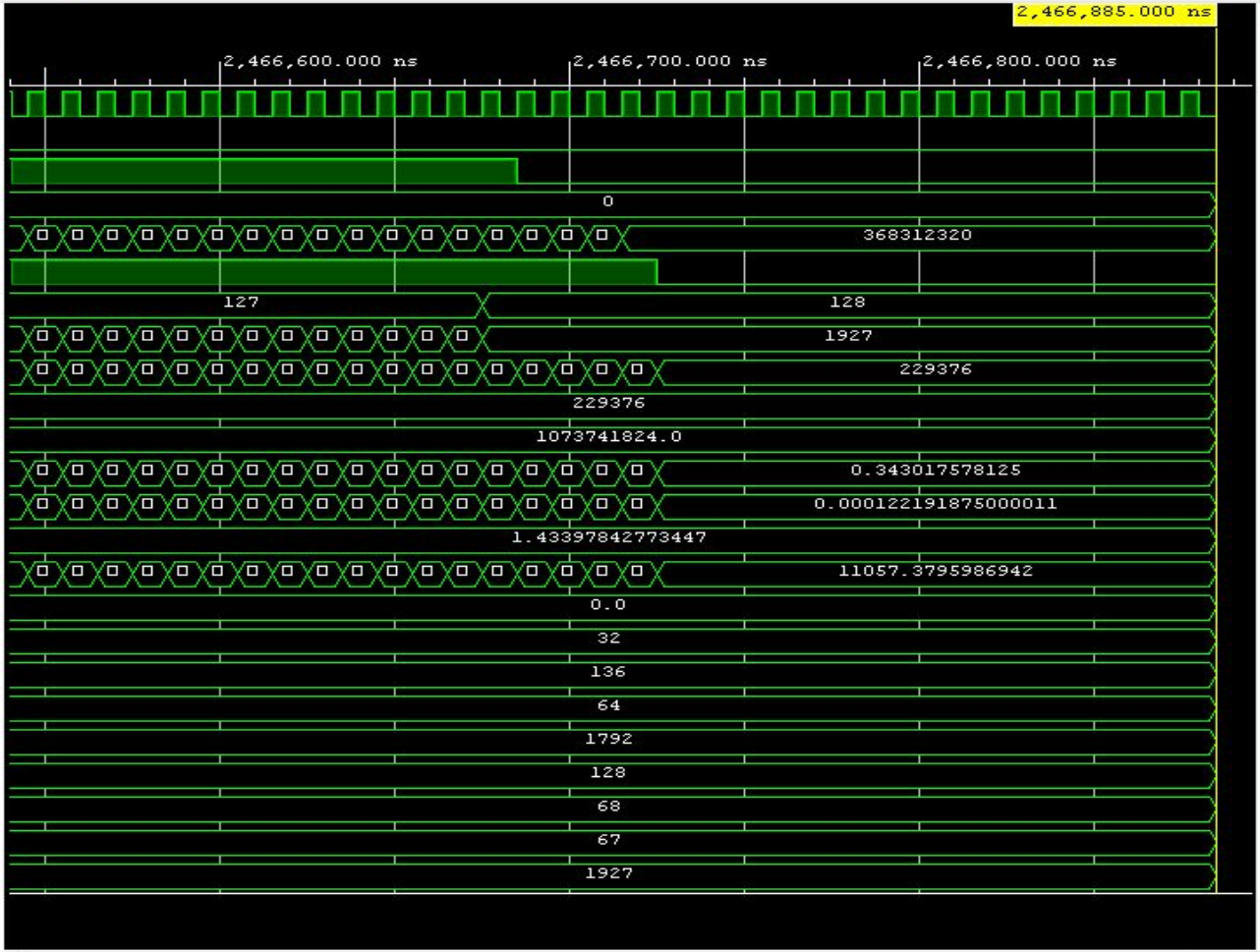
tb.v × rtl.v × Untitled 4 ×

Name	Value
dout[63:0]	368312320
dout_valid	0
curr_row[31:0]	128
curr_col[31:0]	1927
out_idx[31:0]	229376
i[31:0]	229376
scale_out	1073741824.0
rtl_float_val	0.343017578125
error	0.000122191875000011
max_error	1.43397842773447
DATA_WIDTH[31:0]	32
TAPS[31:0]	136
ACCUM_WIDTH[31:0]	64
N_ROWS_X[31:0]	1792
N_COLS_X[31:0]	128
PAD_LEFT[31:0]	68
PAD_RIGHT[31:0]	67
ROW_LENGTH[31:0]	1927



tb.v | Untitled 2

Name	Value
clk	0
rst	0
din_valid	0
din[31:0]	0
dout[63:0]	368312320
dout_valid	0
curr_row[31:0]	128
curr_col[31:0]	1927
out_idx[31:0]	229376
i[31:0]	229376
scale_out	1073741824.0
rtl_float_val	0.343017578125
error	0.000122191875000011
max_error	1.43397842773447
sum_error	11057.3795986942
mean_error	0.0482063493944188
DATA_WIDTH[31:0]	32
TAPS[31:0]	136
ACCUM_WIDTH[31:0]	64
N_ROWS_X[31:0]	1792
N_COLS_X[31:0]	128
PAD_LEFT[31:0]	68
PAD_RIGHT[31:0]	67
ROW_LENGTH[31:0]	1927



Synthesis using Nexys4

Sources | Netlist

- matched_filter_1d
  - Nets (148)
    - col_count (16)
    - data0 (15)
    - dout (64)
    - <const0>
    - <const1>
    - clk
    - clk_IBUF
    - clk_IBUF_BUFG
    - col_count[0]_i_2_n_0
    - col_count[0]_i_3_n_0
    - col_count[15]_i_3_n_0
    - col_count[15]_i_4_n_0

Net Properties

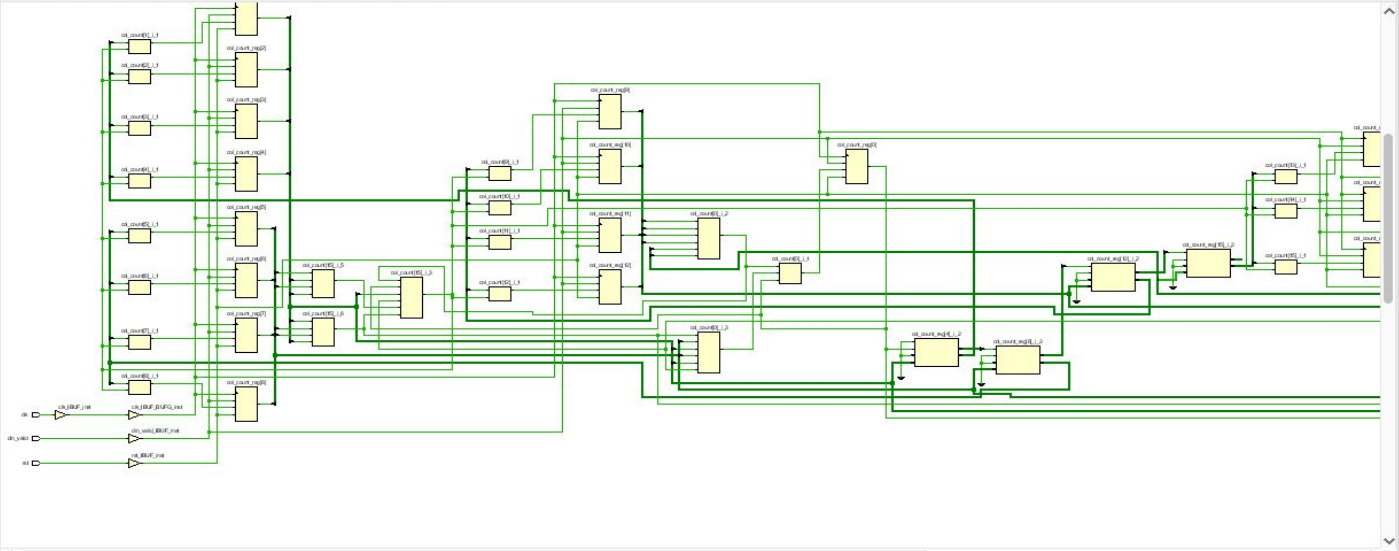
pipe_valid_2

Name:	pipe_valid_2
Type:	SIGNAL
Route status:	Has unplaced ports or pins

General | Properties | Connectivity | Power | Aliases

Project Summary | Device | conv.v | tb.v | Schematic

118 Cells | 100 I/O Ports | 148 Nets



Tcl Console | Messages | Log | Reports | Design Runs | Find Results

Name	Direction	Board Part Pin	Board Part Interface	Interface	Neg Diff Pair	Package Pin	Fixed	Bank	I/O Std	Vcco	Vref	Drive Strength	Slew Type	Pull Type	Off-Chip Termination
rst	IN								default (LVCMOS18)	1.800				NONE	NONE
din[31]	IN								default (LVCMOS18)	1.800				NONE	NONE
din[30]	IN								default (LVCMOS18)	1.800				NONE	NONE
din[29]	IN								default (LVCMOS18)	1.800				NONE	NONE
din[28]	IN								default (LVCMOS18)	1.800				NONE	NONE
din[27]	IN								default (LVCMOS18)	1.800				NONE	NONE
din[26]	IN								default (LVCMOS18)	1.800				NONE	NONE

Results and error (RTL vs MATLAB)

Root Mean Square Error (RMSE): 0.002022

```
# run 1000ns
Loading memory files...
Streaming transposed data into pipeline...
INFO: [USF-XSim-96] XSim completed. Design snapshot 'tb_matched_filter_ld_behav' loaded.
INFO: [USF-XSim-97] XSim simulation ran for 1000ns
launch_simulation: Time (s): cpu = 00:00:06 ; elapsed = 00:00:07 . Memory (MB): peak = 1207.867 ; gain = 43.555
run -all
Idx:      0 | RTL:   47.58481 | MATLAB:   47.58617 | Err: 0.001360
Idx:      1 | RTL:   13.84897 | MATLAB:   13.84769 | Err: 0.001275
Idx:      2 | RTL:   -1.64700 | MATLAB:   -1.64790 | Err: 0.000897
Idx:      3 | RTL:    3.11829 | MATLAB:    3.12129 | Err: 0.003008
Idx:      4 | RTL:    0.21155 | MATLAB:    0.21130 | Err: 0.000253
Idx:      5 | RTL:  -33.61694 | MATLAB:  -33.62037 | Err: 0.003424
Idx:      6 | RTL:  -38.48331 | MATLAB:  -38.48181 | Err: 0.001497
Idx:      7 | RTL:   41.73016 | MATLAB:   41.73294 | Err: 0.002779
Idx:      8 | RTL:   61.84052 | MATLAB:   61.83813 | Err: 0.002384
Idx:      9 | RTL:  -35.49057 | MATLAB:  -35.49266 | Err: 0.002090
Idx:     10 | RTL:  -51.80991 | MATLAB:  -51.80672 | Err: 0.003182
Idx:     11 | RTL:   29.81146 | MATLAB:   29.81250 | Err: 0.001037
Idx:     12 | RTL:   20.69357 | MATLAB:   20.69003 | Err: 0.003541
Idx:     13 | RTL:  -23.16217 | MATLAB:  -23.16172 | Err: 0.000451
Idx:     14 | RTL:   13.30920 | MATLAB:   13.31202 | Err: 0.002820
Idx:     15 | RTL:   15.51758 | MATLAB:   15.51619 | Err: 0.001393
Idx:     16 | RTL:  -24.39578 | MATLAB:  -24.39711 | Err: 0.001329
Idx:     17 | RTL:  -14.19461 | MATLAB:  -14.19369 | Err: 0.000923
Idx:     18 | RTL:  -11.80945 | MATLAB:  -11.80914 | Err: 0.000304
Idx:     19 | RTL:   16.52377 | MATLAB:   16.52405 | Err: 0.000281
```

Results and error (RTL vs MATLAB)

```
Verified 10000 pixels...
Verified 20000 pixels...
Verified 30000 pixels...
Verified 40000 pixels...
Verified 50000 pixels...
Verified 60000 pixels...
Verified 70000 pixels...
Verified 80000 pixels...
Verified 90000 pixels...
Verified 100000 pixels...
Verified 110000 pixels...
Verified 120000 pixels...
Verified 130000 pixels...
Verified 140000 pixels...
Verified 150000 pixels...
Verified 160000 pixels...
Verified 170000 pixels...
Verified 180000 pixels...
Verified 190000 pixels...
Verified 200000 pixels...
Verified 210000 pixels...
Verified 220000 pixels...
============================================================
SIMULATION COMPLETE
Total Pixels Verified: 229376
MAX OVERALL HARDWARE ERROR:  1.433978
MEAN OVERALL HARDWARE ERROR: 0.048206
============================================================
$finish called at time : 2466885 ns : File "C:/Users/Student/conv/conv.srcs/sim_1/new/tb.v" Line 105
run: Time (s): cpu = 00:00:04 ; elapsed = 00:00:08 . Memory (MB): peak = 1207.867 ; gain = 0.000
```

Future Scope

- Integrating all the functions blocks together
- Optimisation of the developed pipeline to reduce overall error.
- SVD implementation after the beamforming function.
- Simulation of the data on Vivado and FPGA implementation.